NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY

8875/02

Paper 2 Core Paper

15 September 2014

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
Total	

This document consists of **12** printed pages and **no** blank page.

[Turn over]

Section A

Answer **all** the questions in this section.

1 **Fig. 1.1** shows the effect of increasing substrate concentration on the rate of a particular reaction in the presence and absence of an enzyme.

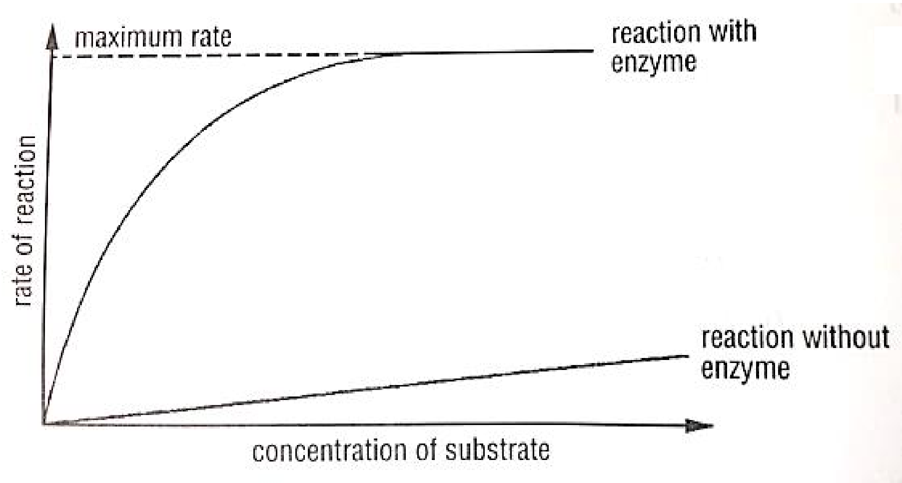

Fig. 1.1

(a) With reference to **Fig. 1.1**, explain the difference between the two graphs.

....

....

....

....

....

.... [3]

(b) On **Fig. 1.1**, draw **two** labelled curves to show the effect on the rate of the enzyme catalysed reaction of the addition of

(i) a competitive inhibitor

(ii) a non-competitive inhibitor

[2]

(c) Explain the effect of a competitive inhibitor on the rate of enzyme activity.

....

....

....

....

....

.... [3]

(d) **Fig. 1.2** below shows an electron micrograph of a cell from the lung that is specialized to take up material by endocytosis. Darkly staining vesicles containing inorganic aluminium silicate crystals are labeled in the figure. These crystals are only ever seen in such cells in the lungs of cigarette smokers. Aluminium silicate is known to be present in tobacco smoke.

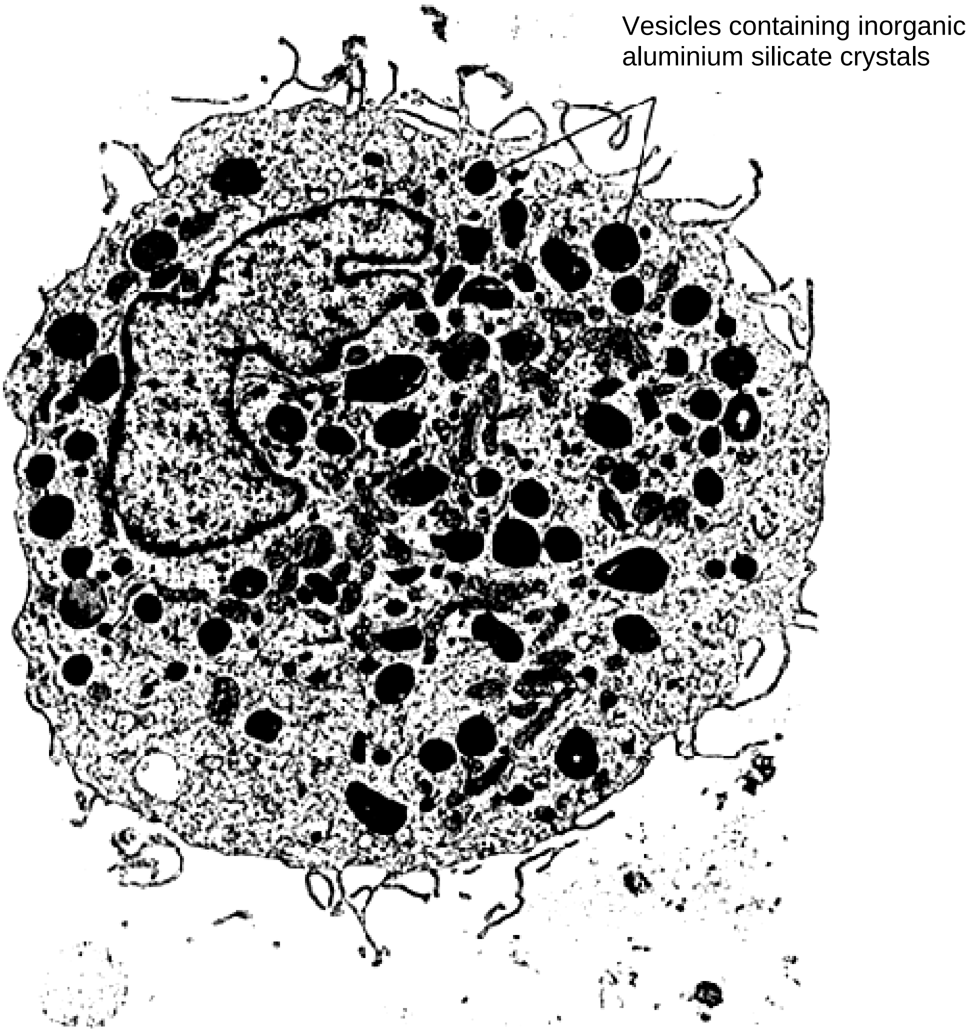

Fig. 1.2

(i) Explain how endocytosis allows foreign material such as aluminium silicate to be taken up by cells.

[2]

(ii) Describe how foreign material is normally dealt with once in these vesicles.

[3]

(iii) Explain the function of the numerous mitochondria that are clearly visible in the cell.

[2]

[Total: 15]

2 A pure-breeding variety of tomato plant, variety **A**, produced red fruit which remained green at their bases even when ripe.

Plants of variety **A** were crossed with another pure-breeding variety, **B**, with orange fruit which have no green bases when ripe. The F_1 generation plants all had red fruit with green bases.

(a) Describe the interaction of the alleles,

(i) at the locus **G**/**g**, controlling green-based or not green-based fruit

....

.... [1]

(ii) at the locus **R**/**r**, controlling red or orange fruit.

....

.... [1]

(b) Using the symbols given in **(a)**, state the genotype of variety **B.**

....

.... [1]

(c) Plants from the F_1 generation were crossed to variety **B** and the offspring were recorded.

red fruit, green base	53
red fruit, non-green base	58
orange fruit, green base	52
orange fruit, non-green base	55

Using the symbols given in **(a)**, draw a genetic diagram to explain the results shown above.

[4]

[Total: 7]

3 Figure **3.1** shows the main stages in the polymerase chain reaction (PCR).

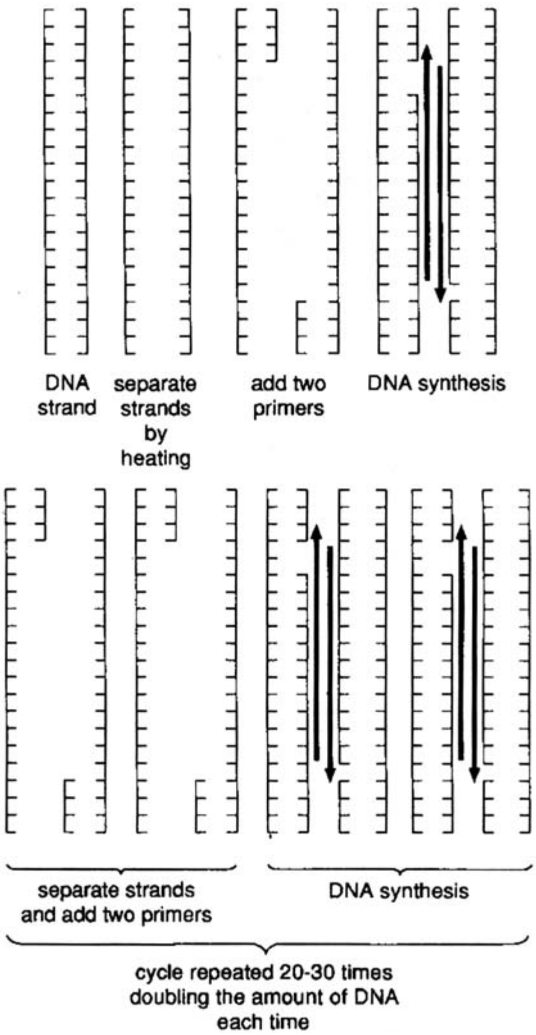

Fig. 3.1

This technique allows multiple copies of a selected length of DNA to be made in the laboratory. Portions of the base sequences at either end of the DNA to be copied are used to produce to short lengths of synthetic DNA, one complementary to each strand of the DNA double helix. These short lengths of DNA serve as primers for *in vitro* DNA synthesis, which is catalysed by DNA polymerase, and they determine the ends of the final DNA fragment that is obtained.

Each cycle of the reaction requires a brief heat treatment to separate the two strands of the DNA double helix. A subsequent cooling of the DNA in the presence of a large excess of the two primers allows their specific attachment to their complementary DNA sequences. The mixture is then incubated with DNA polymerase and the four different DNA nucleotides so that the regions of DNA, starting from the primers, are selectively synthesised.

When first developed, PCR was slow and laborious because a crucial enzyme for the process, DNA polymerase, was denatured at high temperature and has to be added to each cycle. However the introduction of enzyme, *Taq* DNA polymerase (isolated from the heat tolerant bacterium, *Thermus aquaticus*) revolutionised PCR so that the technique could be fully automated.

(a) **(i)** List the components of a nucleotide.

....

.... [1]

(ii) Explain what is meant by *four different nucleotides*.

....

.... [1]

(b) Suggest why the brief heat treatment separates the two strands of the DNA double helix.

....

....

....

.... [2]

(c) Suggest why DNA synthesis takes place in opposite directions on the two separated strands of the DNA double helix.

....

....

....

.... [2]

(d) Explain the advantages of using *Taq* polymerase.

....

....

....

.... [2]

(e) State the limitations of PCR.

....

....

....

.... [2]

[Total: 10]

4 In 1999, genetically modified maize was widely grown in the maize-growing areas of the USA. One of the genetically modified varieties of maize contains a gene (*Bt*) from a bacterium, *Bacillus thuringiensis*. The gene codes for a toxin, which is expressed in the leaves and acts as an insecticide.

(a) Explain briefly

(i) what is meant by a **genetically modified** variety of maize;

....

.... [1]

(ii) why farmers in the USA grow maize carrying the *Bt* gene.

....

....

....

.... [2]

The *Bt* gene is also expressed in pollen, which may be dispersed at least 60m away from the parent plant, by the wind. In the USA, milkweed frequently grows around the edge of maize fields and is fed upon by caterpillars of the Monarch butterfly.

In an experiment into the environmental effects of *Bt* maize, leaves of milkweed were divided into three groups, **A**, **B** and **C**, and treated in the following ways:

A dusted with pollen from genetically modified maize carrying the *Bt* gene;
B dusted with pollen from maize that had not been genetically modified;
C not dusted with pollen.

Monarch caterpillars were then placed on the leaves and the mean area of leaf eaten per caterpillar and the survival of the caterpillars was measured over four days. The results of the experiment are shown in **Fig**. **4.2** and **4.3**.

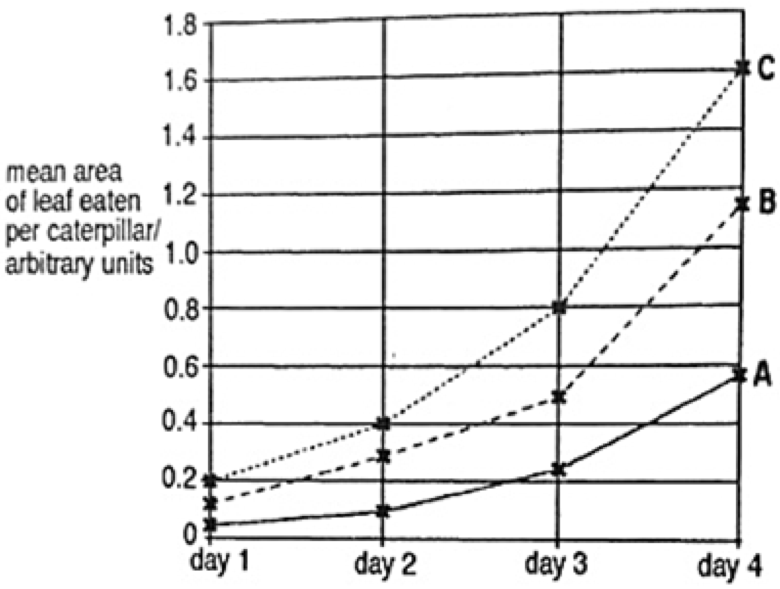

Fig. 4.2

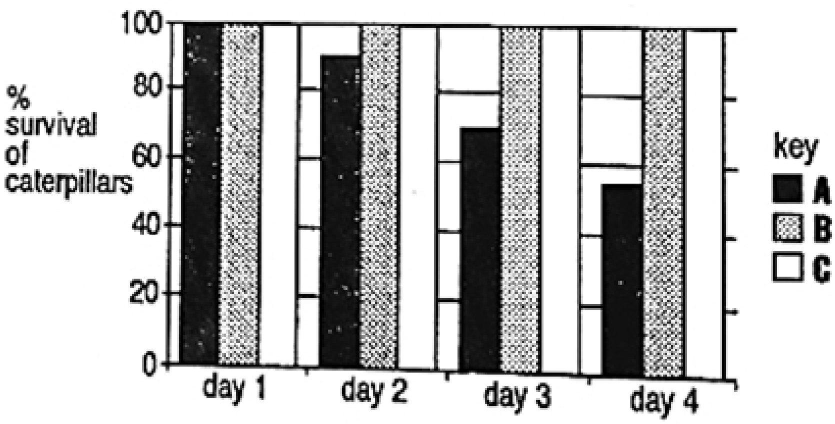

Fig. 4.3

(b) With reference to **Fig. 4.2** and **4.3**, comment on the effect of eating pollen from different maize plants on

(i) the mean area of leaf eaten per caterpillar

[2]

(ii) the survival of Monarch butterfly caterpillars

[3]

[Total : 8]

Section B

Answer **ONE** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Explain how genetic variation can arise. [7]

(b) Explain using two named examples, how environmental forces act as forces of natural selection. [7]

(c) Discuss the benefits and ethical concerns of the human genome project for humans. [6]

[Total: 20]

6 **(a)** Distinguish between the structure of protein and DNA. [7]

(b) Describe the properties of plasmids that allow them to be used as DNA cloning vectors. [5]

(c) With reference to the term ‘limiting factor’, discuss the effect of temperature on the rate of photosynthesis. [8]

[Total: 20]